

TUTORIAL – 8

- Q1.** Describe SSL Handshake protocol with client authentication feature
- Q2.** In SSL, how are the MAC Keys, Encryption Keys and Initial Vectors (IVs) derived from pre_master_secret?

SSL/TLS Handshake

CLIENT



3 Verify Server Cert

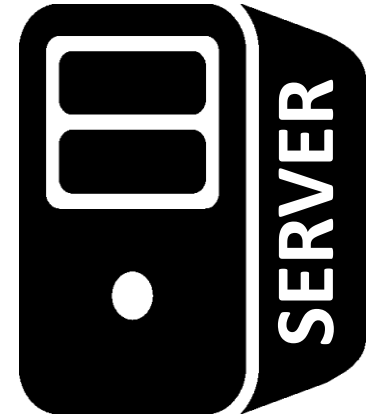
1 "Client Hello" with Cryptographic Info

2 "Server Hello" with Certificate, CipherSuites (& Cert Req)

4 Sends pre-master key (and client Cert)

6 "Finished" by both

7 start exchanging message



5 Verify Client Cert

① Agree Protocol Version

② Select Cryptographic Algorithms.

③ Mutually Authenticate by certificates

④ Use PKE for shared key for SYMM encryption

 **SSL (Secure Sockets Layer) predecessor of TLS**

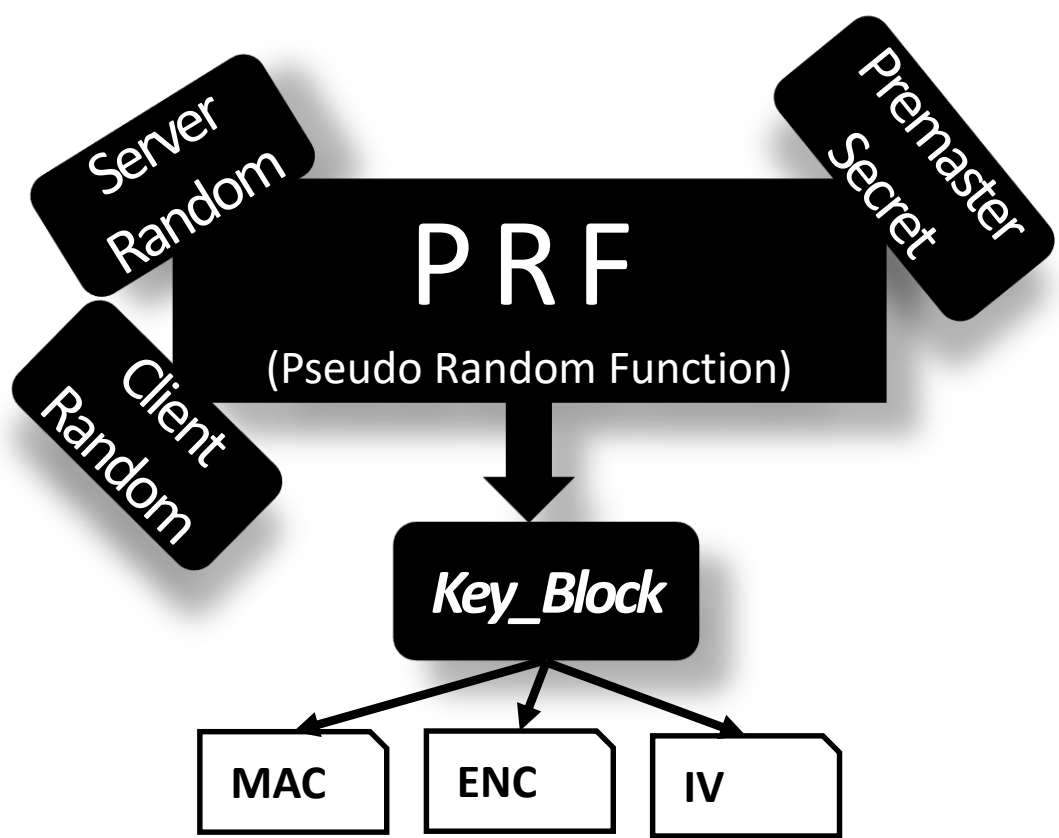
 **Standard to establish security between client & server**

PRE-MASTER SECRET



Master_Secret = PRF(pre_master_secret, ← 1
"master secret", ← 2
ClientHello.random ← 3
+ ServerHello.random) ← 4

MAC KEY, ENC KEY & IV Generaton



```
key_block = PRF(SecurityParameters.master_secret,  
                "key expansion",  
                SecurityParameters.server_random +  
                SecurityParameters.client_random);
```